Note

We note that $(x_i - \bar{x}) \geq 0$ for all observations x_i, $i = 1,2,3,\ldots,n$. If the deviations from the mean $(x_i - \bar{x})$ are small, then the squares of the deviations will be very small.

8.2.4 Variance

The mean of the squares of the deviations from the mean is called Variance. It is denoted by σ^2 (read as sigma square).

$$\text{Variance} = \text{Mean of squares of deviations}$$

$$= \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \ldots + (x_n - \bar{x})^2}{n}$$

$$\text{Variance } \sigma^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}$$

Thinking Corner

Can variance be negative?

8.2.5 Standard Deviation

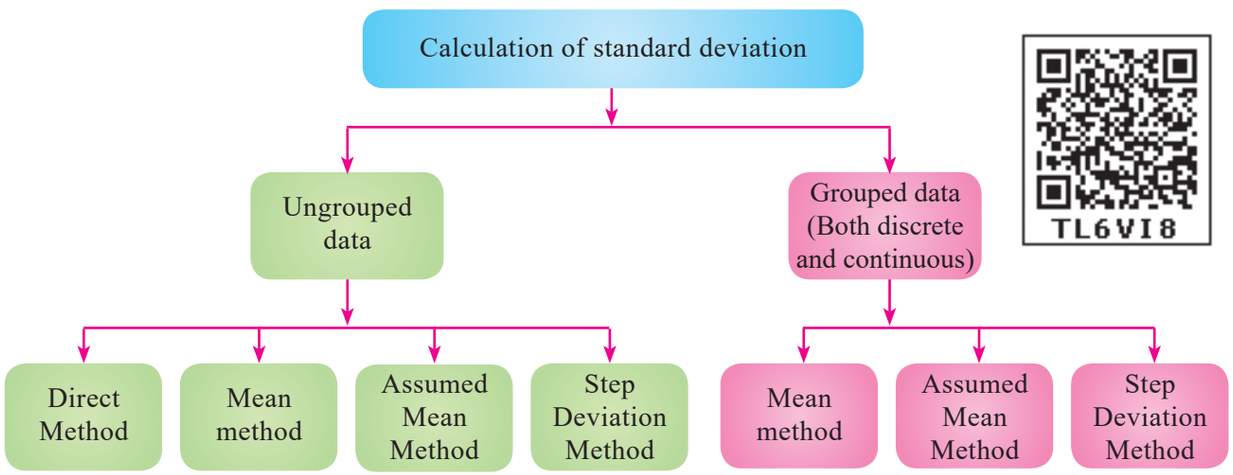
The positive square root of Variance is called Standard deviation. That is, standard deviation is the positive square root of the mean of the squares of deviations of the given values from their mean. It is denoted by σ.

Standard deviation gives a clear idea about how far the values are spreading or deviating from the mean.

$$\text{Standard deviation } \sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}}$$

Do You Know?

Karl Pearson was the first person to use the word standard deviation. German mathematician Gauss used the word Mean error.



TL6VI8

Calculation of Standard Deviation for ungrouped data

(i) Direct Method

$$\text{Standard deviation } \sigma = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n}}$$

$$= \sqrt{\frac{\Sigma(x_i^2 - 2x_i\bar{x} + \bar{x}^2)}{n}}$$

Note

The standard deviation and mean have same units in which the data are given.

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